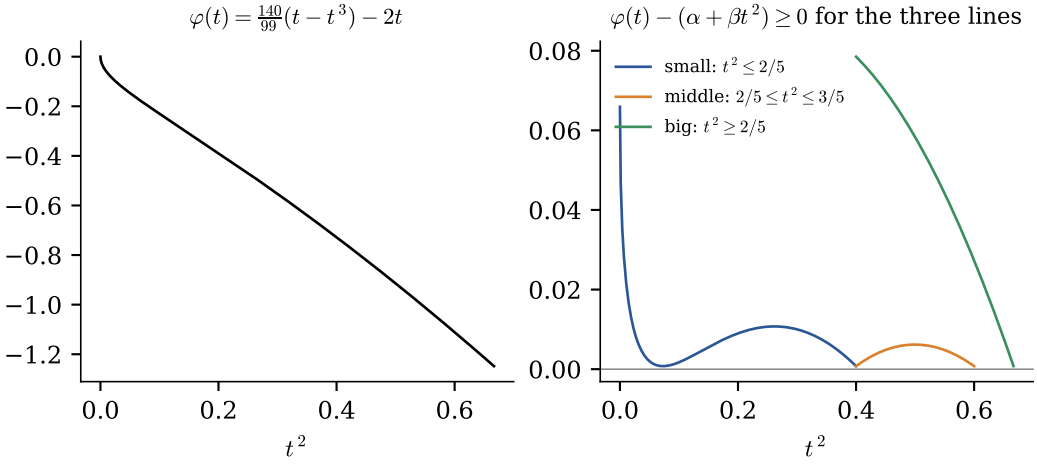
$\varphi(t) = \frac{140}{99}(t - t^3) - 2t$
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.2
0.4
0.6
$t^2$
$\varphi(t) - (\alpha + \beta t^2) \geq 0$ for the three lines
small: $t^2 \leq 2/5$
middle: $2/5 \leq t^2 \leq 3/5$
big: $t^2 \geq 2/5$
0.08
0.06
0.04
0.02
0.00
0.0
0.2
0.4
0.6
$t^2$